IPC Standards Overview

IPC-2221 / 7351 / 6012 / A-610 / J-STD-001 — the rules everyone follows

Industry standards that govern PCB design, footprints, fab, and assembly

Audience: Junior Engineers

What IPC is • Key standards • Class 1/2/3 • Compliance • How to use them

What is IPC?

The trade association for electronic interconnect

- IPC = Institute for Printed Circuits (renamed: Association Connecting Electronics Industries)
- Founded 1957 — based in Bannockburn, IL, USA
- Publishes 200+ standards covering PCB design, materials, fabrication, assembly, test
- Standards are NOT legally mandatory but are de-facto required by most major buyers
- Used by: every major contract manufacturer (Foxconn, Jabil, Flex, etc.)
- Standards updated every 3-5 years; current revisions matter (e.g., IPC-A-610 Rev H, J, K)
- Member companies include Apple, Tesla, Boeing, Samsung, every CM, etc.

IPC Quality Classes — 1 / 2 / 3

Different tiers for different end-use products

Class 1 — Consumer

- TVs, toys, hobby boards
- Most permissive standards
- Cosmetic defects OK
- Cheapest assembly
- Most common for low-cost consumer

Class 2 — Industrial / Telecom

- Servers, networking, industrial
- Mid-tier acceptability
- Most contract-manufactured product
- Industry standard if not otherwise specified

IPC Class 3 — High-Reliability

Medical, aerospace, military, automotive safety

- Class 3 = high-reliability — strictest standards
- Used in: medical implants, aerospace, military, automotive safety
- Stricter rules for: solder voids, joint shape, plating thickness, traceability
- Higher inspection requirements (X-ray on all BGAs, etc.)
- Cost premium: 30-100% over Class 2
- Documentation: full traceability of every component lot
- Class 3 fab + Class 3 CM required — not all manufacturers qualify
- If your buyer demands Class 3, this affects EVERY step of the supply chain

Most-Used IPC Standards

What every PCB designer should know

Standard	Topic	Used by
IPC-2221	Generic Standard on Printed Board Design	All PCB designers
IPC-2222	Sectional Design Standard for Rigid Boards	Standard rigid PCB
IPC-2223	Sectional Design Standard for Flex Boards	Flex / rigid-flex designers
IPC-7351	Generic Requirements for Surface-Mount Land Patterns	All landpattern creation
IPC-6012	Qualification and Performance Spec for Rigid PCBs	PCB fabs + buyers
IPC-6013	Qualification and Performance Spec for Flex PCBs	Flex fabs + buyers
IPC-A-600	Acceptability of Printed Boards (visual + electrical)	Incoming inspection
IPC-A-610	Acceptability of Electronic Assemblies (PCBA)	Assembly inspectors
J-STD-001	Requirements for Soldered Electrical Assemblies	Solder process control
IPC-7711/21	Rework and Modification of Electronic Assemblies	Rework technicians
IPC-2152	Current Carrying Capacity in Printed Boards	Trace-width sizing
IPC-D-275	Solder Mask Specification (older)	Legacy mask reference

IPC-2221 — Generic Design Standard

The PCB design rule book

- Generic rules for ALL PCB types — rigid, flex, single-layer to multilayer
- Conductor width + spacing rules per Class (1 / 2 / 3)
- Clearance rules between conductors at different voltages
- Hole and pad sizing rules
- Solder mask + silkscreen rules
- Component placement guidance
- Combined with IPC-2222 (rigid) or 2223 (flex) for material-specific details
- Most-quoted standard in PCB design reviews and DFM checks

IPC-7351 — Land Pattern Standard

Footprint design — same chip, same landpattern across vendors

- Standardizes the SMD footprint geometry for every package family
- Three density levels:
- • Level A (Most): more land area, easier soldering, less dense
- • Level B (Nominal): balance — industry default
- • Level C (Least): tight, dense, requires precise assembly
- Includes: pad size, pad spacing, courtyard, fiducial position
- Vendor-provided KiCad libraries follow IPC-7351 Level B by default
- Audit your footprints against IPC-7351 — many web-sourced libraries are wrong
- Saturn PCB Toolkit and PCB Libraries Inc. provide IPC-7351 calculators

IPC-6012 — Rigid PCB Qualification

What the fab guarantees about your bare PCB

- Performance + qualification spec for rigid PCBs (single-layer to HDI)
- Defines: dielectric thickness tolerance, Cu thickness, plating, drill quality
- Class 1 / 2 / 3 acceptance criteria
- Microsection inspection: cut a sample, polish, view under microscope
- Acceptance: no copper voids in PTH walls, no laminate cracks, etc.
- Required test coupons in panel for impedance + microsection
- Most reputable fabs are IPC-6012 Class 2 certified by default
- Class 3 fabs are fewer, more expensive — required for medical / aerospace

IPC-A-610 — Assembly Acceptability

The bible of solder joint inspection

- Defines what a 'good' solder joint looks like — and what's defective
- Covers: joint shape, solder amount, copper exposure, gold pad embrittlement
- Photos + diagrams for every defect category
- Class 1 / 2 / 3 acceptance criteria — Class 3 is strictest
- Pass/fail decisions in CM inspection use IPC-A-610
- Latest revision: IPC-A-610 Rev K (2025)
- Inspectors should be IPC-A-610 CIS-certified (instructor) or CIT (technician)
- Required reading for anyone building or buying assembled boards

J-STD-001 — Soldering Process Standard

Process requirements (vs A-610's quality requirements)

- Joint developed with IPC + JEDEC — Joint Standard
- Defines REQUIREMENTS for soldering process (vs IPC-A-610 which defines acceptability)
- Materials: solder alloy, flux type
- Process: temperature profile, time-above-liquidus
- Workmanship: lead trim, joint cleanliness, ESD handling
- Class 1 / 2 / 3 process strictness
- Latest revision: J-STD-001 Rev H (2020)
- Operators should be J-STD-001 certified (CIS instructor or CIT technician)

IPC-2152 — Current Carrying Capacity

How wide your trace needs to be for current

- Defines trace width vs current vs temperature rise
- Replaces older IPC-2221 trace-current charts (which were too conservative)
- Inputs: Cu weight, trace width, ambient temp, allowed temp rise
- Outputs: max safe current for those parameters
- Includes inner-layer vs outer-layer derating
- Software available (IPC-2152 calculator) — paid
- Free alternatives: KiCad calculator, Saturn PCB Toolkit, online JLCPCB calc
- Bottom line: use a calculator, not eyeball, for any trace > 1 A

How to Use IPC at Work

Practical application

- Quote IPC standards in design specs (e.g., 'IPC-A-610 Class 2')
- Reference IPC in fab quote terms
- Use IPC-7351 footprints from libraries
- Cite IPC-A-610 in inspector training
- Audit DFM against IPC-2221 generic rules
- Update specs when revisions release

- Buy the standards (₹15-50k per std on IPC website)
- Many CMs / fabs provide PDF access to customers
- Use as defence in disputes (fab vs designer)
- Refer to IPC certified solderers vs IPC-A-610 inspectors
- Track revisions: IPC-A-610 Rev K, J-STD-001 Rev H, etc.
- Cite specific section numbers in DFM reviews

How to Get IPC Standards

Cost + access options

- Direct from IPC: ipc.org — ₹15-50k per standard for non-members
- IPC member companies get discounts (~50% off)
- Many CMs (Foxconn, Jabil) include relevant standards in their customer portal
- Engineering libraries at major universities often have IPC standards
- Trade subscriptions (e.g., IEEE Xplore) sometimes include IPC
- Indian IPC affiliates: ELCINA, MAIT (smaller libraries)
- Free alternatives: IPC's free 'IPC Edge' video courses for newer engineers
- Be wary of pirated copies — non-current revisions cause real harm

Related Non-IPC Standards

What IPC doesn't cover

AS 9100

Aerospace QMS
beyond IPC

IATF 16949

Automotive QMS
required by car OEMs

ISO 13485

Medical device QMS

MIL-STD-883

Military electronics testing

IPC + JEDEC

Joint IC packaging standards

UL / IEC

Safety + EMC compliance

Key Takeaways

Key takeaways from this lecture

IPC = electronics rulebook

200+ standards used globally.

Class 1/2/3 tiers

Pick by end use.

IPC-7351 for footprints

Source-of-truth.

IPC-A-610 for assembly QC

Defines acceptable joints.

J-STD-001 for process

Solder workmanship.

Quote standards in specs

Anchor your design + reviews.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC official site — ipc.org
- IPC standards listing + buying
- IPC certifications (CIS, CIT) for solderers/inspectors
- ELCINA / MAIT India for local access
- Major CM customer portals (Foxconn, Jabil, Flex)
- IPC Edge — IPC's free video learning platform
- Phil's Lab YouTube — IPC standard walkthroughs
- ASME / IEEE engineering library partnerships

Thank you • Build something this week.